JNTUH SEP 08-05-17PM Set 2

R13

Code No: 118BJ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, May - 2017
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) What is an embedded system? [2]
b) Explain the different applications of embedded system. [3]
c) What are the advantages of Assembly language? [2]
d) Explain signed subtraction with example. [3]
e) Define task and Task state. [2]
f) What is meant by concurrency of task execution in real time system? [3]
g) What is the necessity of RTOS? [2]
h) Explain difference applications of queues. [3]
i) Write the specifications of SHARC. [2]
j) What is network's importance in embedded system? [3]

PART - B

(50 Marks)

2.a) Explain the challenges in embedded computing system design.
b) Explain how to interface external memory to 8051? [5+5]

OR

3. With neat sketch, explain the architecture of 8051 microcontroller. [10]

4. Write a subroutine which checks the content of 20H. If it is a positive number, the subroutine. Find its two's complement and store it in same location and returns. [10]

OR

5.a) Explain the different types of bit-level and byte-level jump instructions in 8051.
b) Classify the arithmetic operations and explain any five. [5+5]

6.a) Explain how interfacing can be done for A/D converter to 8051.
b) Explain how serial data communication will be done in 8051? [5+5]

OR

7.a) Write about message queue management function calls in RTOS.
b) Explain Interrupt Routines in an RTOS Environment. [5+5]

8.a) Explain with an example how semaphores solve the shared data problem.

b) Discuss various methods adopted to reduce power consumption in embedded applications. [5+5]

OR

9.a) Explain the differences between a 'Host Computer System' and a 'Target System' in terms of their hardware and software.

b) Why are most designers use FOSS tools in embedded system development? Discuss. [5+5]

10. Explain block diagram of ARM architecture with neat diagram. [10]

OR

11. Explain the operation of the CAN bus at the physical layer and data link layer with the help of CAN data frame format diagram. [10]

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